

LP / LPB



LP – Value Line

LP – Belt Drive



LP - Value Line / LP - Belt Drive

Operating Manual



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1.1 Service Contact

Please contact our customer service department if you have any technical questions:

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Fax: +49/ (0) 79 31 / 493-903

E-Mail: service@alphagetriebe.de

Address:

alpha getriebebau GmbH

Walter-Wittenstein-Str. 1

D-97999 Igersheim

2 General Information

2.1 Description, Designations

The low-backlash planetary gear reducer LP (subsequently termed gear reducer) is constructed in the "M" (motor-mounted) version.

The LPB version consists of a low-backlash planetary gear reducer and a toothed belt pulley (optional). It is designed for feed drives.

2.2 Whom does this manual concern?

This manual concerns all persons who install, operate, or maintain this gear reducer.

They may only carry out work on the gear reducer, if they have read and understood this operating manual. Please pass the safety instructions on to other persons as well.

2.3 Which signs and symbols are referred to in this manual?

➡ An "action instruction", which requires you to carry out an action.

▽ With a "check" you can specify whether the device is ready for the next work stage.

😊 A "usage tip" shows you an option of facilitating or improving operations.

The safety instructions symbols are described in the "Safety" section.

2.4 Exclusion of liability

The manufacturer does not accept liability for damage or injury ensuing from improper handling of the gear reducer.

2.5 Modifications, Reconstructions

Modifications or reconstructions of the gear reducer may only be carried out with the express written authorisation of **alpha getriebebau**.

2.6 EC Machinery Directive

Within terms of the EC machinery directive 98/37 EC, the gear reducer is not considered an autonomous machine, but as a component to install in machines.

Operation is prohibited within the area of validity of the EC directive, until it has been determined that the machine, in which this product is installed, corresponds to the regulations within this directive.

2.7 Technical Modifications

alpha getriebebau reserves the right of carrying out technical modifications to improve the product.

2.8 Copyright

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3 Safety

3.1 Intended Use

The gear reducer is designed for industrial applications. Please refer to our catalogue or our Internet page for the maximum permitted torques and speeds:

www.alphagetriebe.de/gear.

- ➡ Please consult our service department (see [1.1](#)), if your gear reducer is older than a year. In this way you receive valid data.

3.2 Improper Use

Every usage which exceeds the limits stated above (especially higher torques and speeds) is not compliant with the regulations, and is thus prohibited.

The operation of the gear reducer is prohibited if:

- it was not mounted according to regulations (e.g., securing the motor),
- it was not installed according to regulations (e.g., securing screws),
- the gear reducer is very soiled,
- it is operated without lubricant.

3.3 Safety Instructions

The following symbols are used in this operating manual to warn you of hazards:



DANGER!

This symbol warns you of danger of injury to yourself and others.



Attention

This symbol warns you of the risk of damage to the gear reducer.



Environment

This symbol warns of environmental pollution risk.

3.3.1 General Safety Instructions

Working on the gear reducer



DANGER!

Inappropriately executed work can lead to injury and damage.

- ➡ Make sure that the gear reducer is only installed, maintained, and dismantled by trained technicians.



DANGER!

Foreign bodies spinning through the air can cause grave injury.

- ➡ Before putting the gear reducer into operation, check that there are no foreign bodies or tools near the gear reducer.



Attention

Loose or overloaded screwed connections can cause damage.

- ➡ Tighten and check all screwed connections for which a tightening torque is specified, on principle with a calibrated torque wrench.

Operation



DANGER!

Touching hot surfaces can lead to burns.

- ➡ Do not touch the gear reducers if their operating temperatures are too high, or use suitable safety equipment (e.g. gloves).



DANGER!

Rotating machinery can lead to injury. There is danger of being trapped or pulled in!

- ➡ Keep a sufficient distance to rotating machinery.

Maintenance**DANGER!**

An unintentional start of the machine during maintenance work can lead to serious accidents.
 ➔ Make sure no one can start the machine while you are working on it.

**DANGER!**

Even a brief running of the machine during maintenance work can lead to accidents if the safety devices are not operating.
 ➔ Make sure that all safety devices are mounted and active.

Lubricants**DANGER!**

Extended, intensive contact with synthetic oils can lead to skin irritations.
 Avoid extended contact with oil, and clean oil off skin thoroughly.

**DANGER!**

Hot oil may cause scalding.
 When changing oil, protect yourself against contacting hot oil.

**Attention**

Mixing different lubricants can impair the lubricant properties. This can destroy the gear reducer.
 ➔ Only re-fill with the lubricant type that is in the gear reducer.
 ➔ If you wish to use another lubricant, carry out a complete oil change (with flushing).

**Environment**

Lubricants (oils and greases) are hazardous substances, which can contaminate soil and water.
 Collect drained lubricant into suitable receptacles and dispose of it according to the valid national guidelines.

3.4 In case of fire

The gear reducer itself is non-flammable. However, the interior of the standard version has mineral oil-based lubrication.

➔ Please observe the following instructions if the gear reducer is situated in a burning environment.

3.4.1 Suitable extinguishing agents, Protective equipment

Carbon dioxide, powder, foam, fog

**DANGER!**

High temperatures produce irritating steam.
 ➔ Use a protective breathing apparatus.

3.4.2 Unsuitable extinguishing agents

Do not spray with water!

3.4.3 Additional Information**Environment**

➔ Prevent the penetration of the lubricant into drains, sewers, and water resources.
 Further information on the OPTIMOL LONGTIME PD 1 lubricant can be ordered from:
 Castrol Industrie GmbH,
 Munich, Germany,
 Tel.: +49 (0) 89 / 41 83-116

4 Technical Specifications

4.1 Design

4.1.1 Motor-mounted “M”

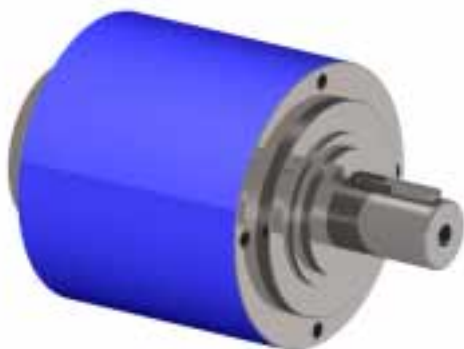


Fig. 4.1

The clamping hub enables a quick and easy mounting of the motor:
The motor is centred using the clamping hub and not with the adapter plate. The motor can thus be mounted without radial distortion.
The reducer can be adapted to various motors using the adapter plate and spacer sleeve. The gear reducer has been designed to compensate for thermal linear expansion of the motor shaft.

4.1.2 Toothed belt pulley “B” option

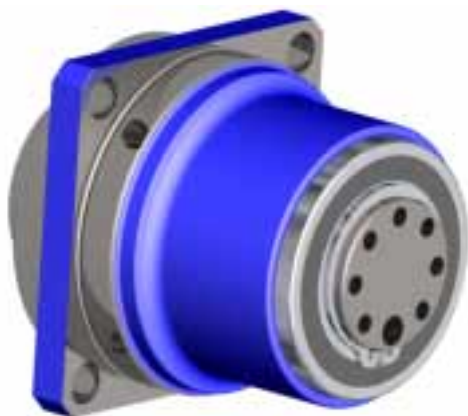


Fig. 4.2

The specifications for mounting the motor also apply to the LPB gear reducer. However, the LPB gear reducers have an output flange instead of an output shaft. You thus have the option of mounting a toothed belt pulley. The respective assembly dimensions can be found in the LP / LPB product catalogue.

4.2 Weight

The weight of the gear reducer ranges from 0.77 to 21 kg.

☺ The [Table 5.1](#) in section 5.2 helps you in a more exact specification of the weights.

4.3 Lubricant quantities and types



Attention

No changing of lubricant for LP and LPB gear reducers!

➡ Never open the housing of these gear reducers.

The gear reducers have been filled with lubrication and sealed during manufacture. They are lubricated for their entire working life.

The ambient temperature may not be under -10 °C and not over $+40\text{ °C}$. Operating temperature may not exceed $+90\text{ °C}$.

Divergent operating conditions may make different lubricant quantities and different lubricants necessary.

➡ In these cases, please consult **alpha getriebebau** (see section [1.1](#)).

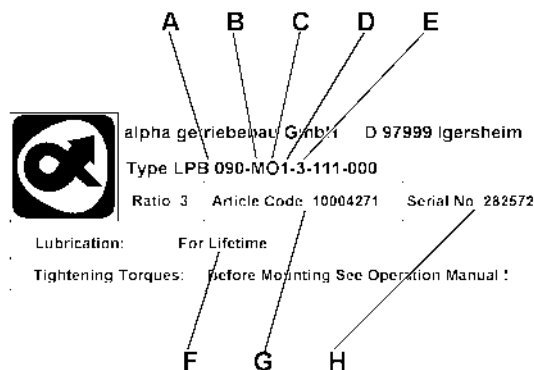
4.4 Performance Statistics

Please refer to our catalogue or our Internet page for the maximum permitted torques and speeds: <http://www.alphagetriebe.de/gear>.

- ➡ Please consult our service department (see [1.1](#)) if your gear reducer is older than a year. In this way you receive valid data.

4.5 Identification Plate, Ordering Key

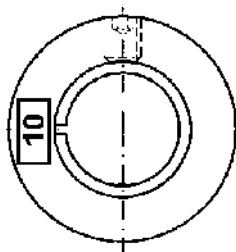
The identification or type plate is located on the gear reducer housing.



The following specifications can be found on the type plate:

- A Gear reducer type:**
LP: 050/070/090/120/155
LPB: 070/090/120
- B Gear reducer versions:**
M = motor-mounted gear reducer
- C Gear reducer version:**
0 = Standard version
X = Special gear reducer
- D Number of stages:**
1 = 1-stage
2 = 2-stage
- E Ratio: e.g. 3**
- F Lubrication note:**
e.g. For lifetime
- G Article code: e.g. 10004271**
- H Serial number: e.g. 282572**

Fig. 4.3



The ratio (e.g. $i = 10$) can also be found on a label on the plug socket. The spacer sleeve is properly aligned when the slit points to the label.

Fig. 4.4

5 Delivery Status, Transport, Storage

5.1 Delivery Status

Within Europe, the gear reducers are packed in cardboard boxes with paper padding. The paper padding is re-usable and can be recycled.

To ship overseas, the gear reducers are wrapped in foil (PE) and foamed into the cardboard box (diphenylmethane). Please dispose of packing material according to the valid national regulations.

All gear reducers are treated with an anti-corrosion agent at the gear input and output. All gear reducers are filled with lubricant during manufacture.

The adapter plate for mounting the motor is securely mounted only on LP gear reducers.

Of special note for LPB gear reducers

When a tooth belt drive and adapter plate are ordered, these arrive with mountings that have **only been tightened by hand**. The final mounting needs to be carried out by the customer (see section [6.3.2](#) and [6.4.1](#))

5.2 Transport

No special direction or position is prescribed to transport the gear reducer.

The tables shown below are designed to help you specify the weights of your gear reducer.

Weight [kg]					
Size	050	070	090	120	155
Version					
LP 1-stage	0.77	1.9	4.1	9.0	17.5
LP 2-stage	0.95	2.2	5.1	11.2	21.0
LPB 1-stage	-	1.5	3.4	7.4	-

Table 5.1

5.2.1 Transport using hoisting equipment



DANGER!

Falling loads or breakage of fastening equipment can cause injury.

- ➔ Do not stand under suspended loads.
- ➔ Keep as safe a distance as possible from securing equipment.



Attention

Falling or hard placement can damage the gear reducer.

- ➔ Only use hoisting and securing equipment which is permitted for the size / weight of your gear reducer.
- ➔ Ensure that the load is slowly and carefully handled and placed.

5.3 Storage

The gear reducer can be stored dry and in a horizontal position in the original packing for a maximum of 2 years at a temperature between 0°C and +30°C. As storage logistic, we recommend the “first in - first out” principle.

6 Mounting, Putting into Operation

- ➔ Please observe the instructions in the [„General Safety Instructions”](#) in section [3.3.1](#).

6.1 Preparing mounting

All gear reducers are treated with an anti-corrosion agent at the gear input and output.

Remove all traces of the anti-corrosion agent before mounting the gear reducer.



Attention

Pressurised air can damage the gear reducer seals, and thus lead to leakage.

- ➔ Do not blow out the flanges and shafts with pressurised air when cleaning.

Four threaded bores are available in the gear unit housing for bolting it to your machine (refer to Table 6.1.1 and Table 6.1.2). In the LPB gear reducers are four through-holes.

LP: Threaded bores in the gear unit housing			
Gear reducer size	Bore Ø [mm]	For bolt size / Property class	Tightening torque [Nm]
LP 050	44	M4 / 12.9	4.95
LP 070	62	M5 / 12.9	9.7
LP 090	80	M6 / 12.9	16.5
LP 120	108	M8 / 12.9	40
LP 155	140	M10 / 12.9	81

Table 6.1.1

LPB: Through-holes in gear unit housing				
Gear reducer size	Bore Ø [mm]	Quantity x Diameter [] x [mm]	For bolt size / Property class	Tightening torque [Nm]
LPB 070	82	4 x 9	M8 / 12.9	40
LPB 090	106	4 x 11	M10 / 12.9	81
LPB 120	144	4 x 13	M12 / 12.9	140

Table 6.1.2

Of special note for LPB gear reducers

When a tooth belt drive, adapter plate and motor are ordered, these arrive with mountings that have **only been tightened by hand**. The final mounting needs to be carried out by the customer.

- ➔ Loosen the motor, adapter plate and the toothed belt pulley and lay ready all the component parts for the final assembly.

If necessary, a spacer may need to be placed between gear reducer and your machine for the precise positioning (measure L2 in product catalogue) of the gear reducer (see also [Fig. 6.2](#)). This spacer is **not** included in the gear reducer's scope of delivery.

The bolts for mounting the motor to the adapter plate depend on the motor and need to be provided by the customer.

- ➔ Observe the changed order of steps for mounting the LPB gear reducer: Only in this case must you attach the gear reducer to your machine before you attach the motor. (When doing so, also observe the instructions in sections [6.2](#), [6.3](#) and [6.4](#))

6.2 Mounting the LPB gear reducer onto your machine

The information in section 6.2 is only applicable for the LPB gear reducer.

- ➔ Please observe the instructions in the [„General Safety Instructions“](#) in section [3.3.1](#).
- ➔ Thoroughly clean the output flange, centring, and fitting surface.

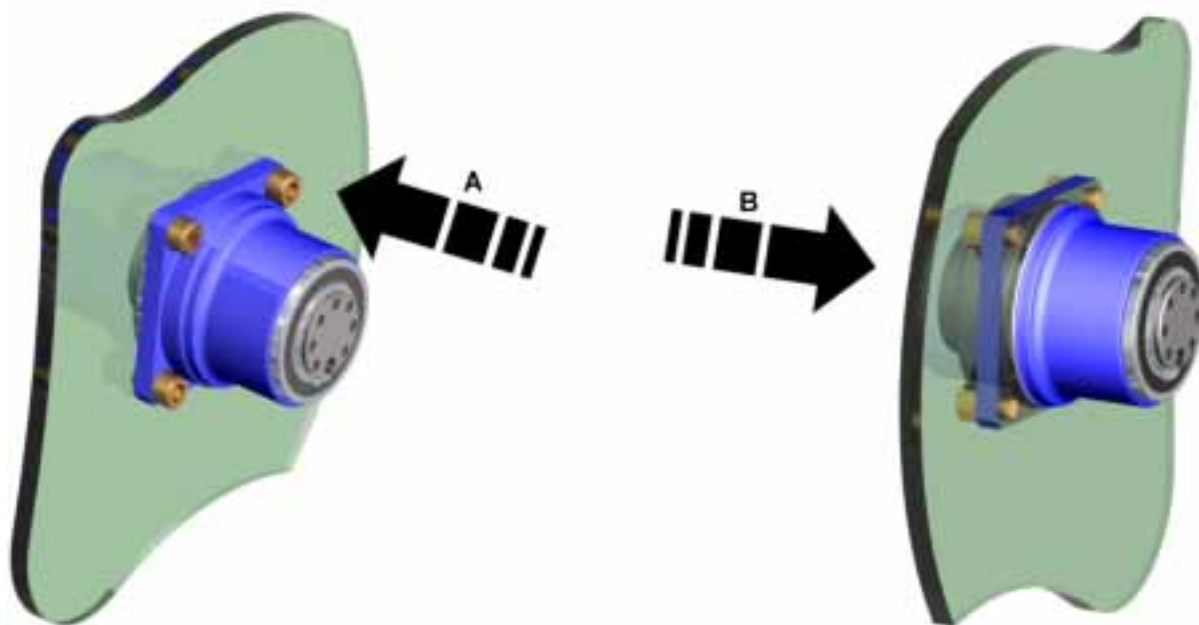


Fig. 6.1

There are two centring mechanisms for assembling the gear reducer:

- A Outer centring
- B Inner centring

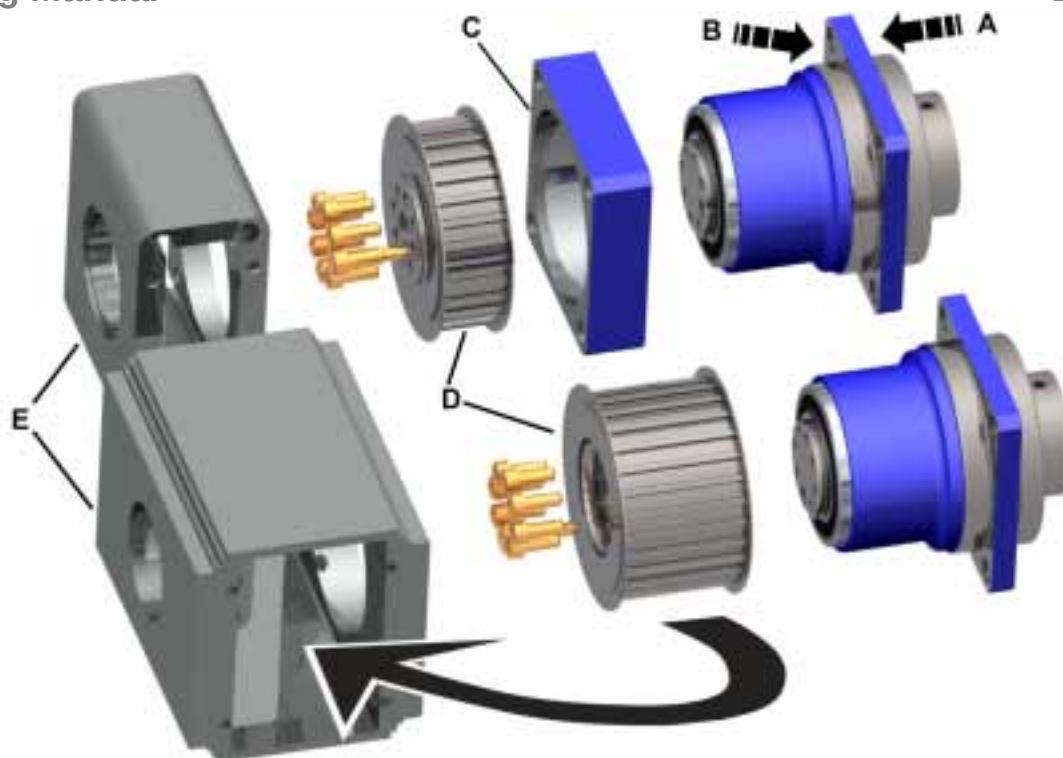


Fig. 6.2

- ☺ When using hollow profiles: Position the toothed belt pulley (D) in the hollow profile (E) before you attach the gear reducer.
- ☺ Mount the gear reducer in such a way that the type plate remains legible. There may be threaded holes or through-holes provided in your machine for attaching the gear reducer. If necessary, a spacer (C) may need to be placed between gear reducer and your machine to position the gear reducer precisely. This spacer is not included in the gear reducer's scope of delivery.
- ☺ You can find the prescribed screw sizes in [Table 6.1.2](#).
- ▽ Check whether the given tightening torques and screw depths are suitable for your machine's material (e.g. aluminium, cast iron, steel).
- ➡ Coat the four screws with screw-bonding agent (e.g. Loctite 243) and screw the gear unit housing and your machine together.

6.3 Motor-mounting

6.3.1 General information on motor mounting

Only for standard gear reducers of the LP series, which are delivered with motor attached, is the motor already tightly mounted. The further instructions for this in section 6.3 may therefore be ignored.

If LPB gear reducers are delivered with motor, the motor is only tightened by hand. For these as well as for all gear reducers delivered without a motor, the following applies:

The gear reducer is provided for the motor mounting.

The motor to be mounted must:

- correspond to the B5 design,
- show a radial and axial runout tolerance of "N" according to DIN 42955 and
- if possible, have a smooth shaft.



Attention

Distortion can damage the motor and the gear reducer.

- ➡ Ensure that the motor is mounted in a vertical position.

6.3.2 Pre-mounting adapter plate

In standard gear reducers of the LP series, the adapter plate is already mounted.

The information in section 6.3.2 is only applicable for the LPB gear reducer.

The adapter plate along with the four fastening bolts are included in the scope of delivery.

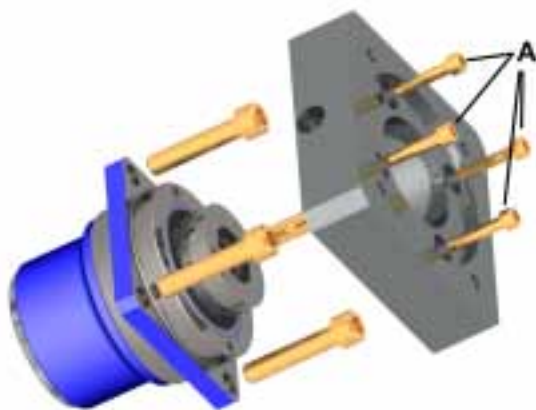


Fig. 6.3

- Place the adapter plate on the housing and tighten the bolts (A) initially by hand.
- Tighten the bolts (A) in diagonal order making at least two passes to the required tightening torque (refer to Table 6.2).

Fastening bolts adapter plate		
Gear reducer size	Size	Tightening torque [Nm]
LPB 070	M4	2.9
LPB 090	M6	9.9
LPB 120	M6	9.9

Table 6.2

6.3.3 Tools for tightening the clamping hub

The threaded pin can be tightened through the mounting bores of the adapter plate using a square 1/4 inch socket spanner. In addition a torque wrench is needed for the respective torque range.

6.3.4 Mounting the motor

- If the motor shaft has a feather key, remove the feather key. If recommended by the motor manufacturer, apply a half wedge.

A plug socket and a slit spacer sleeve connect the motor shaft and the gear drive shaft.

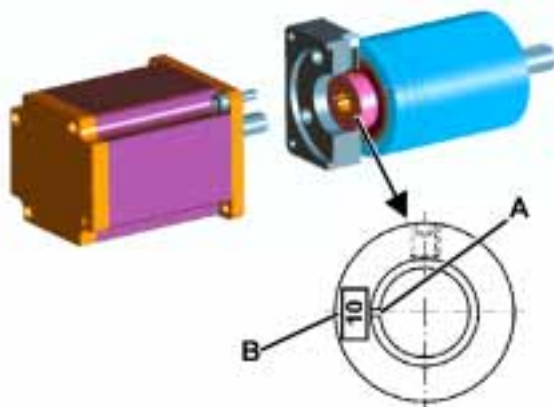


Fig. 6.4 shows an LP gear reducer for clarity's sake.

- Clean the plane surfaces of the motor and drive.
- Clean the motor shaft, the plug socket bore hole, and the spacer sleeve.
- Turn the plug socket so that the threaded pin can be reached through the mounting holes in the adapter plate.
- Take care that the slot of the spacer sleeve is positioned 90° to the threaded pin.
- ▽ The spacer sleeve is properly aligned when the slit (A) points to the label (B).
- Readjust the spacer sleeve if necessary.



Attention

Excessively high axial forces can damage the motor and drive.

- Ensure that the axial forces that occur are not higher than the values specified in Table 6.3.

Gear reducer size	Max. Axial force [N]
LP 050	45
LP(B) 070	80
LP(B) 090	100
LP(B) 120	150
LP 155, 1-stage	180
LP 155, 2-stage	150

Table 6.3

**Attention**

Motors with

- shaft shoulder,
- distinctive chamfer radius, or
- longer shafts than are permitted for the relevant drive

lead to distortions in mounting, which damage the motor and the drive.

- ➔ Check the interfering edges by measuring, or by a measurement check based on our catalogue specifications and the information of the motor manufacturer.
- ➔ Please consult our service department to obtain a wider adapter plate or an intermediary flange.

**Attention**

Distortion can damage the motor and the drive.

- ➔ Ensure that the motor is mounted in a vertical position.



Fig. 6.5 shows an LP gear reducer for clarity's sake.

- ➔ Position the motor so that the plane surfaces fit together.
- ▽ Make sure that the motor allows itself to be moved into position "easily".
- ▽ There is to be no gap between the motor, drive and adapter plate.
- ➔ Coat the bolts with screw-bonding agent (e.g. Loctite 243), and screw the motor to the adapter plate.
- ➔ Tighten the threaded pin to the tightening torque (Table 6.4).

Threaded pin motor assembly		
Gear reducer size	Tightening torque [Nm]	Width across flats [mm]
LP 050	5.6	3
LP(B) 070	9.5	4
LP(B) 090	23	5
LP(B) 120	45	6
LP 155, 1-stage	78	8
LP 155, 2-stage	45	6

Table 6.4

- ➔ Press the enclosed stopper plugs up to their stops in the mounting bores of the adapter plate.

6.4 Mountings on the gear output side

- ➔ Thoroughly clean the output flange or shaft, centring, and fitting surface.

6.4.1 Mountings on the LPB output flange

The information in section 6.4.1 is **only** applicable for LPB gear reducers.

**Attention**

Distortions during mounting operations can damage the drive.

- ➔ Mount toothed belt pulleys onto the output flange without forcing.
- ➔ Do not on any account attempt an assembly by force or hammering.
- ➔ Only use suitable tools and equipment.

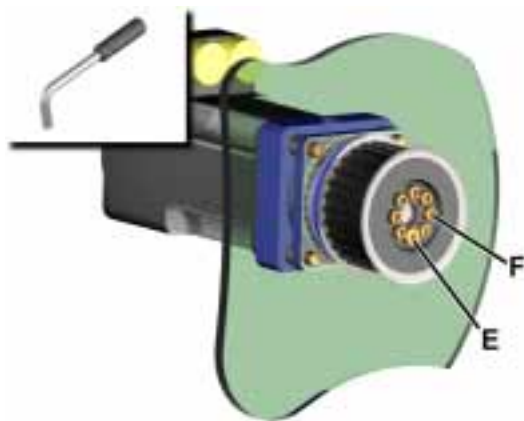


Fig. 6.6

- Coat the contact surfaces lightly with grease or lubricant in order to prevent contact corrosion.
- Place the toothed belt pulley onto the output flange and tighten the bolts initially by hand.
- Tighten the fastening bolts (F) and the close-tolerance bolt (E) (1 piece) in diagonal order making at least two passes to the required tightening torque.

Toothed belt pulley				
Gear reducer size	Close-tolerance bolt (property class 12.9)		Fastening bolts (property class 12.9)	
	Size	Tightening torque [Nm]	Quantity x size	Tightening torque [Nm]
LPB 070	M5x12	9.7	5 x M5x12	9.7
LPB 090	M6x16	16.5	7 x M6x16	16.5
LPB 120	M6x12	16.5	7 x M8x20	40

Table 6.5



The tension of the tooth belt is brought about by its construction. The toothed belt's prestressing force influences the ball bearing life of the drive.

- You can determine the theoretical bearing life for each case using our analysis software Cymex 2.1.
- Set your toothed belt so that there are no lateral starting loads pressing on the toothed belt pulley.

Fig. 6.7

6.4.2 Mountings on the output shaft



Attention

Distortions during mounting operations can damage the gear reducer.

- Mount gearwheels and toothed belt pulleys onto the output shaft without forcing.
- Do not on any account attempt an assembly by force or hammering.
- Only use suitable tools and equipment.
- When shrink-fitting a gear onto the output shaft: Ensure that the maximum static axial forces of the output bearing (Table 6.6) are not exceeded.

Gear reducer size	LP 050	LP 070	LP 090	LP 120	LP 155
F_{amax} [N]	1800	4300	5100	11300	18500

$$S_0 = 1.8 \quad F_r = 0$$

Table 6.6

6.5 Mounting the gear reducer onto your machine

The information in section 6.5 is only applicable for the standard gear reducer (without LPB).

- Please observe the instructions in the [„General Safety Instructions“](#) in section [3.3.1](#).
- ☺ Mount the gear reducer in such a way that the type plate remains legible.
- Coat the four screws with screw-bonding agent (e.g. Loctite 243), and screw the gear unit housing and your machine together.
- ☺ You can find the prescribed screw sizes and tightening torques in [Table 6.1.1](#) and [Table 6.1.2](#).

6.6 Putting into operation

- Please observe the instructions in the „[General Safety Instructions](#)“ in section [3.3.1](#).

7 Operation

7.1 Operating conditions

- Please observe the instructions in the „[General Safety Instructions](#)“ in section [3.3.1](#).

The gear reducer must be installed in a clean and dry environment. Coarse dust and liquids of all kinds impair its function.

The specifications for lubricants and operating temperatures can be found in section [4.3](#). Divergent operating conditions may make different lubricant quantities and different lubricants necessary.

- In these cases, please consult **alpha getriebebau**.

8 Maintenance

8.1 Shutdown, preparation

- Please observe the instructions in the „[General Safety Instructions](#)“ in section [3.3.1](#).
- Shut down the machine in which the gear reducer is installed.
- Disconnect the machine from the mains, before starting maintenance work.

8.2 Inspection schedule

Maintenance work / See section...	Maintenance periods			
	At Start-up	After 500 oper- ating hours or 3 months	Every 3 months	Yearly
Visual Inspection / 8.3.1	X	X	X	
Checking the tightening torques / 8.3.2	X	X		X

Table 8.1



Attention

No changing of lubricant for LP and LPB gear reducers!

- Never open the housing of these gear reducers.

The gear reducers have been filled with lubrication and sealed during manufacture. They are lubricated for their entire working life.

8.3 Maintenance Work

8.3.1 Visual Inspection

- Check the entire gear reducer by carrying out a thorough visual inspection for exterior damage and oil leakage.

8.3.2 Checking the tightening torques

- Check the tightening torques of the fastening bolts on the drive housing. For LPB gear reducers, also check the fastening bolts on the toothed belt pulley.
- ☺ You can find the prescribed tightening torques in [Table 6.1.1](#), [Table 6.1.2](#) and [Table 6.5](#) in section 6.
- Check the tightening torque of the threaded pin on the motor mounting.
- ☺ You can find the prescribed tightening torques in [Table 6.4](#) in section 6.

8.4 Start-up after maintenance work

- ➡ Clean the outside of the gear reducer.
- ➡ Assemble all safety devices.
- ➡ Do a test run, before re-releasing the machine for operation.

8.5 Malfunction list (troubleshooting)

- ➡ Seek an immediate solution if you notice oil loss, increased noise during operation or higher operating temperatures.

Malfunction	Possible cause	Solution
Higher operating temperature	Dimensioning insufficient	Check the technical data
	Motor is heating the gear reducer	Check the wiring of the motor, replace the motor or install insulation between motor and gear reducer
	Ambient temperature too high	Ensure adequate cooling
Increased noise during operation	Damaged bearings	Please consult our customer service department.
	Damaged gear teeth	
Loss of lubricant	Lubricant quantity too high	Wipe off discharged lubricant and continue to watch the gear reducer. Lubricant discharge must stop after a short time.
	Sealings not tight	Please consult our customer service department.

Table 8.2

9 Supplementary Information

- ☺ If you should need supplementary information (e.g. disassembly, or disposal), please contact our service department (chapter [1.1](#))

10 Appendix

10.1 Tightening torques

property class	Tightening torque [Nm] for threads...													
	M 3	M 3.5	M 4	M 5	M 6	M 8	M 10	M 12	M 14	M 16	M 18	M 20	M 22	M 24
4.8	0.56	0.86	1.28	2.5	4.3	10.5	21	36	58	88	121	171	230	295
8.8	1.28	1.96	2.9	5.75	9.9	24	48	83	132	200	275	390	530	675
10.9	1.8	2.75	4.1	8.1	14	34	67	117	185	285	390	550	745	950
12.9	2.15	3.3	4.95	9.7	16.5	40	81	140	220	340	470	660	890	1140

Table 10.1

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